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The Editors
Information and Computation

Re: Submission of a regular research article — “The Rate–Distortion Theory of Bounded Sequential Transduction: A Comparative Syntax for Finite-State Approximation”

Dear Editors,

I am submitting the manuscript referenced above for consideration as a regular research article in *Information and Computation*. The manuscript develops a rate–distortion theory of bounded sequential transduction in which the rate is a state budget M and the distortion is a regime-specific task loss, and it spans precisely the journal’s stated scope: theoretical computer science together with computational applications of information theory. The work is original, has not been published previously, and is not under consideration by any other journal. The submission comprises the manuscript (238 pages, 18 sections, 390 theorem-like environments of which 141 carry proofs, with a 39-entry bibliography and a back-matter notation index) and a supplementary package, both described below.

What the paper establishes. The manuscript gives one comparative syntax for finite-state approximation across three regimes that until now have been studied with disjoint instruments. Commitment targets deterministic specifications through Mealy machines and Myhill–Nerode residuals; retention targets stochastic controlled processes through stationary controlled unifilar causal machines and their lumpable quotients; grounding targets stochastic channels through linear finite-rank Hankel realizations, whose resource is operator rank rather than quotient index. All three share one syntactic core — histories, finite-index right congruences, quotient machines, stationary aggregation, and variational gaps — while their semantic objects differ, and every approximation claim carries an explicit type signature fixing regime, feasible set, aggregation, support, protocol, and analytical bridge, so that statements from different regimes cannot be silently conflated.

The contributions we would foreground. First, the comparative syntax itself, together with its negative results: an independence theorem shows that the three regimes’ approximation obstructions vary independently, so no cross-regime ordering of gaps holds. Second, the type discipline: seven-coordinate signatures that make regime boundaries mechanically checkable, which is what allows the comparative statements to be stated at all. Third, the determination-index ladder $(\kappa_{\text{pair}}, \kappa_{\text{obs}}, \kappa_{\text{det}})$, which organizes commitment-side approximability; exact-phrase searches over arXiv and the open web (2026-09-02) confirm the terms enter the literature unclaimed. Fourth, the protocol stratification: mistake complexity $\Theta(M \log M)$ in the reset-word, persistent-stream, and active residual-identification protocols, with agnostic regret $\Theta(\sqrt{TM \log M})$. On the retention side the paper also proves an exact information-bottleneck decomposition, NP-completeness in the Gaussian quadratic restricted regime, and a sharp spectral converse; on the grounding side it identifies the unrestricted gap with σ_{M+1} by Eckart–Young–Mirsky and a superpolynomial determinism gap.

Positioning. I would state plainly that the per-regime instruments are, for the most part, transports of known tools — information-bottleneck decompositions, spectral converses, singular-

value approximation, minimax lower-bound techniques — into the unified accounting. The contribution is the comparative layer: the shared schema, the independence and non-orderability results, the type discipline, and the exact constants that become meaningful only when the regimes are placed side by side. The manuscript’s introduction positions the work against the automata minimization, computational-mechanics, and Hankel/AAK operator-theoretic literatures, and its coined terms carry disambiguation footnotes where neighboring literatures use similar phrases.

Length and referee navigation. The manuscript is long, and it was engineered for that: a table of contents, a notation index of four cross-referenced tables resolving roughly fifty bespoke symbols to their defining sites, consolidated cross-references, and theorem numbering within sections. The three regimes are individually self-contained chapters, so a referee can navigate by the section map on page one. A preprint of this version will be posted to arXiv concurrent with the submission, to timestamp the coined vocabulary and the results while review proceeds.

The multiletter-AAK status, stated honestly. The most exposed single claim concerns the Hankel-structured grounding gap for multilet alphabets. The manuscript proves a conditional-form theorem: the transport argument is proved, and its conclusion holds under a structure-preserving embedding hypothesis that is recorded as an explicit open problem, not as an established fact. This material has been internally proof-checked to the line level, its supporting computational claims have been independently reproduced, and it has no external corroboration — which is the documented reason it received a dedicated check. The abstract and the theorem’s scope remarks state the conditional status explicitly, so no downstream claim overstates it.

Verification depth. Two dedicated proof-check reports accompany this submission as optional editor material; between them they record line-level checks of all 141 proof-bearing results, including the multiletter-AAK material. The supplementary package converts every quoted computational observation into a runnable check: a 41-check numerical verification suite, the exhaustive enumeration programs with their exact outputs and extremal machine tables, and the Lean 4 development of the machine-checked fragment — seventeen statements across seven modules against Mathlib, shipped with a build script and an axiom-audit gate that recompiles the development and verifies every tracked declaration against Lean’s standard three axioms. The manuscript’s availability statement describes all of this, and the supplementary README documents the three places where reproduction is partial or not re-run, with reasons.

Declarations. The author received no funding for this work and declares no competing interests. Generative-AI assistance in drafting and manuscript preparation is declared in the manuscript’s back matter (“Acknowledgments and Declaration of Generative AI”), where the author’s responsibility for the final content is stated. No empirical dataset was used.

Thank you for considering the manuscript. I would be glad to provide any further material that would help the review.

Sincerely,

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